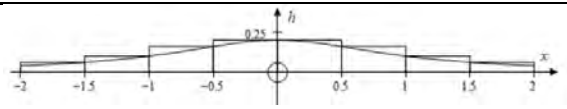
Question			Answer	Marks	AOs		Guidance
8	(a)		 x 0 ±0.5 ±1 ±1.5 2 h $\frac{1}{5}$ $\frac{4}{25}$ $\frac{1}{10}$ $\frac{4}{65}$ $\frac{1}{25}$ h 0.2 0.16 0.1 0.061538 0.04	**M1**	**3.4**	Finding at least 2 distinct values for h Need not be a table of values	
			$=2\times0.5(0.2+0.16+0.1+0.061538)$	**M1**	**1.1a**	Using rectangles forming UB for area with width 0.5. Need not be drawn	Allow both M marks if only half the area considered.
			=0.522 to 3sf	**A1** **[3]**	**1.1**		
8	(b)		Area $\approx\frac{0.5}{2}(0.2+0.04+2(0.16+0.1+0.061538))$ =0.221 (to 3 sf)	**M1** **A1** **[2]**	**1.1a** **1.1**	Using trapezium rule with 5 h values FT incorrect h values for the M mark Allow awrt 0.22	Using the definite integral functionality of calculator gives 0.2214297… So method must be seen.
8	(c)		Volume = $2\times(0.2208\times10)$ $=4.42\text{ m}^3$	**M1** **A1** **[2]**	**1.1a** **1.1**	Condone missing factor of 2 for method mark FT part (b)	